

PHASE 2: SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Coffee Export Value Chain Management System (CEVCMS)

Document Control Information

- **Document Title:** Software Requirements Specification (SRS) for the Coffee Export Value Chain Management System (CEVCMS)
- **Document Reference:** ECTA-CEVCMS-SRS-V2.1
- **Client Organization:** Ethiopian Coffee and Tea Authority (ECTA), Addis Ababa, Ethiopia
- **Prepared By:** Enterprise Systems Architecture & Software Consulting Practice
- **Document Version:** 2.1.0
- **Status:** Final Baseline / Standard Production Release
- **Date:** August 7, 2026

Revision History

Version	Date	Author(s)	Primary Changes / Remarks
0.1.0	May 12, 2026	Lead Requirements Engineer	Initial draft compiled from Phase 1 Discovery Findings.
1.0.0	June 20, 2026	Senior Software Architect	Structural alignment with IEEE 830 / ISO/IEC/IEEE 29148.
2.0.0	July 28, 2026	ECTA Technical Steering Committee	Review integrated; added EUDR compliance & offline sync models.
2.1.0	August 7, 2026	Principal System Architect	Finalized SRS baseline, database schemas, and 40 core use cases.

Approval Sign-Off Matrix

Role	Name	Title / Organization	Signature & Date
Project Sponsor	H.E. Director General	Ethiopian Coffee and Tea Authority	[APPROVED - 07/08/2026]
Lead Business Analyst	Senior Business Analyst	Global Enterprise Consulting Practice	[APPROVED - 07/08/2026]
Software Architect	Lead Software Architect	Enterprise Solutions Division	[APPROVED - 07/08/2026]
ECTA IT Director	Director of IT & Automation	ECTA Technology Division	[APPROVED - 07/08/2026]

Table of Contents

1. [Introduction](#)
2. [Overall Description](#)
 - 2.1 Product Perspective & Integrations
 - 2.2 Product Functions (Module Directory)
 - 2.3 User Classes & Operational Matrix
 - 2.4 Operating Environment
 - 2.5 Constraints
 - 2.6 Assumptions & Dependencies
3. [Functional Requirements](#)
 - FR-MOD-01: Authentication & Identity Management
 - FR-MOD-02: User & Role Management (RBAC)
 - FR-MOD-03: Farmer & Farm Polygon Registry
 - FR-MOD-04: Primary Cooperative & Washing Station Management
 - FR-MOD-05: Coffee Cherry Collection & Batch Intake
 - FR-MOD-06: Processing & Dry Mill Operations
 - FR-MOD-07: Warehouse & Inventory Control
 - FR-MOD-08: Quality Inspection & Q-Grading System
 - FR-MOD-09: Traceability & Dynamic QR Generation
 - FR-MOD-10: GPS Tracking & Digital Waybill Management
 - FR-MOD-11: Export Licensing & Contract Registration
 - FR-MOD-12: Banking, Forex & Financial Settlement
 - FR-MOD-13: Customs Clearance & Port Logistics
 - FR-MOD-14: Exporter & Foreign Buyer Portals

- FR-MOD-15: Notification & Alert Engine
- FR-MOD-16: Analytics, Reporting & Executive Dashboards
- FR-MOD-17: Audit Trail & Compliance Engine
- FR-MOD-18: Offline Synchronization & Local Storage Engine
- 4. [External Interface Requirements](#)
- 5. [Non-Functional Requirements](#)
- 6. [Security Architecture & Requirements](#)
- 7. [Database Requirements & Data Architecture](#)
- 8. [API Specifications & Webhooks](#)
- 9. [Business Rules Engine](#)
- 10. [Comprehensive Use Cases](#)
- 11. [Data Flow Diagrams \(DFD\)](#)
- 12. [Risk Analysis & Mitigation Strategies](#)
- 13. [Testing & Quality Assurance Strategy](#)
- 14. [Deployment Architecture & Infrastructure](#)
- 15. [Future Enhancements & EUDR Roadmap](#)
- 16. [Appendices](#)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional, non-functional, structural, and interface requirements for the **Coffee Export Value Chain Management System (CEVCMS)**. CEVCMS is a national-level enterprise software platform commissioned by the **Ethiopian Coffee and Tea Authority (ECTA)** to digitize, secure, trace, and streamline Ethiopia's coffee export lifecycle—from farm-gate origin to final overseas shipment at port.

This document serves as the binding technical contract between ECTA, software implementation contractors, third-party integration partners (e.g., National Bank of Ethiopia, Customs Commission, Commercial Banks), and system auditors. It strictly adheres to **IEEE Std 830-1998** and **ISO/IEC/IEEE 29148:2018** standards.

1.2 Scope

CEVCMS encompasses an end-to-end digital ecosystem operating across mobile (Android native), web desktop portals, edge offline terminals, and RESTful API microservices. The boundary of CEVCMS spans:

1. **Upstream (Origin):** Farmer registration, GPS farm boundary polygon capturing, primary cooperative cherry collection, weight-scale digital integration, and wet processing lot creation.
2. **Midstream (Processing & Quality):** Secondary dry milling, lot grading at ECTA Liquoring and Inspection Centers, sensory/physical analysis (Q-Grading), dynamic QR code tag generation, and warehouse inventory management.
3. **Downstream (Logistics & Export):** Export contract registration, National Bank of Ethiopia (NBE) foreign exchange (Forex) tracking, Commercial Bank letter of credit (LC) matching, Customs export permits, digital waybill issuing, GPS-monitored transit, container sealing, and port shipment clearance.

4. **Compliance & Analytics:** European Union Deforestation Regulation (EUDR) geolocation compliance, real-time value chain analytics, price discovery engines, automated tax/royalty calculation, and immutable audit logging.

1.3 Business Objectives

- **Abolish Paper-Based Friction:** Convert 100% of physical waybills, grading certificates, and export permits into cryptographically verifiable digital records.
- **Guarantee End-to-End Traceability:** Provide 100% origin verification for exported coffee down to individual farmer plots or primary collection station batches to meet EUDR and global consumer mandates.
- **Eliminate Foreign Exchange Leakage:** Enforce automated cross-validation between export volume, minimum contract pricing, and commercial bank Forex settlement records.
- **Reduce Logistics Lead Time:** Cut average time required for quality grading, waybill issuing, and export clearance from **18 days down to under 48 hours**.
- **Empower Smallholder Farmers:** Guarantee fair pricing transparency through daily SMS/USSD market rate broadcasts and automated weight-scale receipts.

1.4 Project Goals

- Deploy an enterprise multi-tenant cloud core deployed within the Ethiopian Artificial Intelligence Institute (EAI) / Ethio Telecom Tier-III Data Center with zero single points of failure.
- Deploy an offline-first Android application capable of queuing transactions locally in remote rural zones without cellular connectivity for up to 14 consecutive days.
- Establish real-time API connectivity with **8 major external sovereign and commercial ecosystems** within Ethiopia.

1.5 Definitions & Acronyms

Definitions

- **Coffee Batch / Lot:** A unified, trackable quantity of raw, parchment, or clean coffee beans assigned a unique cryptographic Global Identification Number (GIN) and QR tag.
- **Primary Cooperative:** A localized entity collecting raw coffee cherries directly from registered smallholder farmers.
- **Dry Mill:** A processing facility where parchment or dry cherry coffee is hulled, cleaned, sorted, and graded into exportable green coffee.
- **Q-Grading:** The standardized physical and sensory analysis performed by certified ECTA cuppers to assign an official grade (Grade 1 through Grade 9 / Undergrade).
- **Digital Waybill (e-Waybill):** A tamper-evident digital transit permit containing origin, weight, vehicle ID, driver details, route path, and dynamic QR signatures.

Acronyms

- **CEVCMs:** Coffee Export Value Chain Management System
- **ECTA:** Ethiopian Coffee and Tea Authority
- **NBE:** National Bank of Ethiopia
- **ECC:** Ethiopian Customs Commission
- **ECX:** Ethiopia Commodity Exchange

- **EUDR:** European Union Deforestation Regulation
- **GIN:** Global Identification Number
- **RBAC:** Role-Based Access Control
- **HS Code:** Harmonized System Code for international trade
- **USSD:** Unstructured Supplementary Service Data

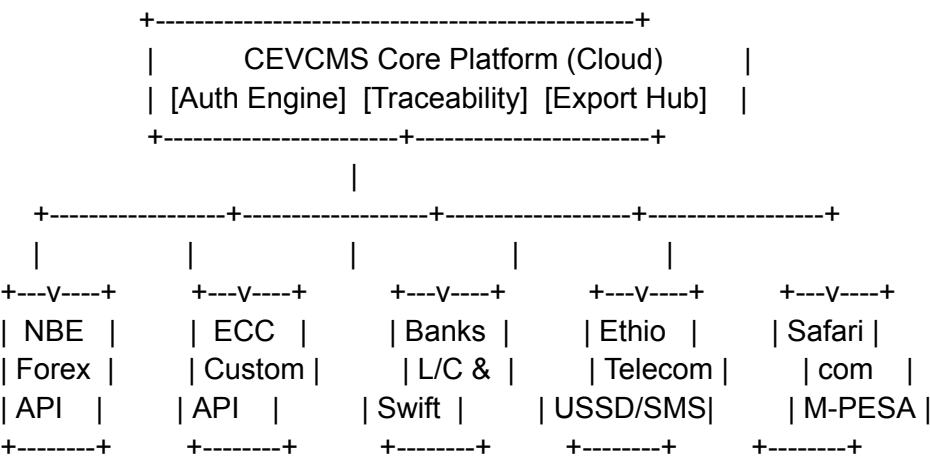
1.6 References

1. *ECTA Phase 1 Discovery Report – Coffee Export Value Chain Digital Transformation* (55 Pages, Baseline Source).
2. ISO/IEC/IEEE 29148:2018 Systems and software engineering — Life cycle processes — Requirements engineering.
3. IEEE Std 830-1998 IEEE Recommended Practice for Software Requirements Specifications.
4. European Union Regulation (EU) 2023/1115 on Deforestation-free Products (EUDR).
5. National Bank of Ethiopia (NBE) Coffee Export Forex Directive No. FXD/82/2023.

2. Overall Description

2.1 Product Perspective & External Integrations

CEVCMS operates as the central digital nervous system of Ethiopia's agricultural export economy. It replaces fragmented desktop spreadsheets, physical paper logs, and isolated departmental databases with a unified microservices ecosystem.



External Integration Interfaces

1. **National Bank of Ethiopia (NBE):** Real-time registration of export contracts, minimum price enforcement, foreign exchange allocation monitoring, and delinquent export tracking.
2. **Ethiopian Customs Commission (ECC):** Automated electronic clearance verification at border posts and ports (Djibouti corridor), matching container seals against registered e-Waybills.
3. **Commercial Banks (CBE, Awash, Dashen, etc.):** Electronic LC notification, advance payment verification, seller-buyer payment settlement, and automatic remittance confirmation.

4. **Ethio Telecom & Safaricom Ethiopia:** Bulk SMS notifications, USSD menu hosting (*899#), direct SIM data connectivity, and M-PESA / Telebirr payment processing for smallholder farm payouts.
5. **Logistics & GPS Provider Gateways:** Ingestion of real-time telematics from GPS transponders attached to coffee transport trucks during transit from regional dry mills to Addis Ababa/Diredawa warehouses and Djibouti port.
6. **Foreign Buyers Portal & Global Roasting Registries:** REST APIs exposing cryptographic traceability records (farm origin, harvest date, cupping score, EUDR compliance hash) to international buyers.

2.2 Product Functions (Module Directory)

Module ID	Module Name	Primary Function	Core Technology
MOD-01	Authentication & Security	Multi-factor login, biometric driver verification, JWT session control.	OAuth2 / OpenID Connect
MOD-02	User & RBAC Management	Enterprise role assignments, granular permissions, division scoping.	Keycloak / Custom RBAC
MOD-03	Farmer & Polygon Registry	Smallholder profiling, GPS farm polygon capture, land title binding.	GeoDjango / PostGIS
MOD-04	Cooperative & Processing	Cherry intake, wet mill batching, moisture recording, farmer payout.	Android Offline / SQLite
MOD-05	Batch & Inventory Control	Warehouse storage, parchment tracking, lot splitting, re-bagging.	Redis / PostgreSQL

MOD-06	Quality & Q-Grading	Physical defect scoring, sensory cupping ratings, grade certificate issue.	Web Canvas / Angular
MOD-07	Traceability & QR Engine	Immutable batch lineage graph generation, QR tag serialization.	Node.js / Crypto Libs
MOD-08	Digital Waybill & GPS	Geofenced transit tracking, driver assignment, e-Waybill issuing.	Leaflet / Traccar API
MOD-09	Export Licensing & Forex	Contract validation, NBE pricing cross-check, export permit generation.	Python FastAPI
MOD-10	Customs & Port Logistics	Container sealing logs, customs officer clearance scanning, port intake.	Native Mobile / Web
MOD-11	Offline Sync Engine	Delta sync, conflict resolution, local cryptographic log validation.	CouchDB / WatermelonDB
MOD-12	Analytics & Reporting	Executive dashboards, export velocity, regional yield prediction.	Apache Superset / Metabase

2.3 User Classes & Operational Matrix

The system explicitly supports **12 primary user classes**.

Role Name	Primary Responsibility	Digital Literacy	Hardware Used	Connectivity	Offline Support
Farmer	Views market rates, receives digital weigh-scale receipts.	Low	USSD / Feature Phone	Cellular / None	N/A (USSD/SMS)
Collection Agent	Registers cherries, weighs intake, issues farmer digital receipts.	Medium-Low	Rugged Android Tablet	Intermittent 2G/3G	Mandatory (14 Days)
Coop Manager	Manages wet mill, oversees batching, initiates transport.	Medium	Android Tablet / Laptop	Intermittent 3G	Mandatory (7 Days)
Dry Mill Operator	Records parchment processing, hulling ratios, output bags.	Medium	Industrial Desktop / PC	Fixed Broadband	Optional (1 Day)
Q-Grader	Conducts blind cupping, scores defects, assigns official grade.	High	Touchscreen Tablet / PC	Fixed Fiber	Optional
Exporter	Registers export contracts, applies for permits, tracks shipments.	High	Desktop Workstation	High-Speed Fiber	No


```

+-----+
| Web Browser Terminals : Google Chrome 115+, Firefox 110+, Edge 115+ |
| Mobile OS Environments : Native Android 10.0+ (API Level 29)         |
| Primary Cloud Hosting : EAll Cloud / Ethio Telecom Tier-III DC      |
| Database Layer        : PostgreSQL 16 with PostGIS 3.4 & Redis 7.2   |
| Container Orchestration: Kubernetes (K8s) cluster with Docker runtime |
| Hardware Interfaces   : Bluetooth Scales, Handheld 2D Scanners, Zebra |
| Telecommunications    : Ethio Telecom / Safaricom USSD/SMS Gateways  |
+-----+

```

2.5 Constraints

1. **Regulatory Compliance:** Must strictly enforce Ethiopian Coffee Export Directives, NBE Forex Regulations, and EUDR (Regulation EU 2023/1115) mandates.
2. **Infrastructure Deficits:** System must function seamlessly in rural regions experiencing power outages up to 72 hours and total network blackout up to 14 days.
3. **Data Sovereignty:** All personal farmer data, financial transactions, and trade volume metrics must reside physically within servers located inside the geographical boundaries of Ethiopia.
4. **Language & Accessibility:** User Interfaces must provide multi-lingual support: **Amharic, Afaan Oromo, Tigrinya, and English.**

2.6 Assumptions & Dependencies

- **Assumptions:** Smallholder farmers have access to at least a basic GSM feature phone capable of receiving SMS or executing USSD strings.
- **Dependencies:** Timely execution of Service Level Agreements (SLAs) with Ethio Telecom and Safaricom for USSD code provisioning (*899#) and dedicated APN tunnels for edge tablets.

3. Functional Requirements

Requirements are categorized by functional module and uniquely identified using the format **FR-[MODULE]-[NUMBER]**.

```

=====
MODULE 01: AUTHENTICATION & ACCESS CONTROL (FR-AUTH)
=====

```

FR-AUTH-001: Multi-Factor Authentication (MFA)

- **Purpose:** Secure platform access across all web and mobile portals.
- **Actors:** All System Users.
- **Inputs:** Username, Password, Time-based OTP (TOTP) or SMS OTP.
- **Outputs:** Authenticated JWT Token Pair (Access Token: 15 min expiration, Refresh Token: 8 hour expiration).
- **Business Rules:** MFA is mandatory for all administrative, bank, customs, and exporter accounts.
- **Validation Rules:** Passwords must meet NIST 800-63B guidelines (min 12 chars, upper/lower/number/symbol).

- **Exception Handling:** Lock account for 30 minutes after 5 consecutive failed login attempts.
- **Priority:** Critical
- **Acceptance Criteria:** User successfully authenticates and receives scoped JWT within 1.2 seconds.

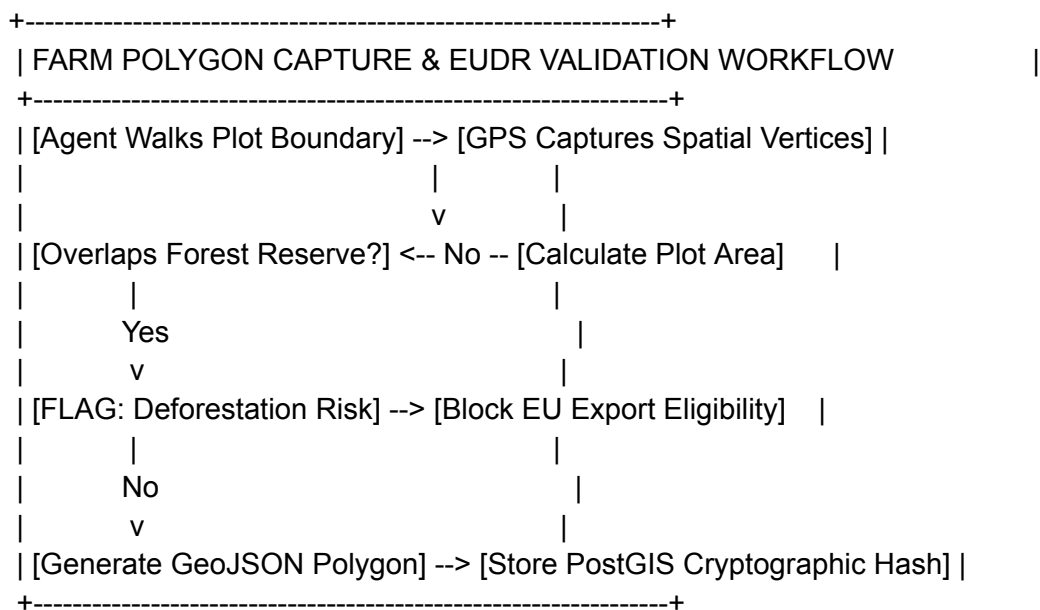
FR-AUTH-002: Biometric Operator Binding (Android)

- **Purpose:** Prevent identity fraud by collection agents and drivers operating shared mobile tablets.
- **Actors:** Collection Agent, Truck Driver.
- **Inputs:** Local Fingerprint scan / Android BiometricPrompt API.
- **Outputs:** Local cryptographic session unlock.
- **Priority:** High

```
=====
MODULE 02: FARMER & FARM POLYGON REGISTRY (FR-FARM)
=====
```

FR-FARM-001: Farmer Master Profiling

- **Purpose:** Establish a single source of truth for smallholder farmers.
- **Actors:** Cooperative Agent, ECTA Field Inspector.
- **Inputs:** Full Name, National ID / Kebele ID, Gender, Household size, Primary Cooperative ID, Phone Number, Bank/Telebirr Account Number.
- **Outputs:** Unique Farmer Identification Number (FIN - Format: ETH-FAR-XXXX-XXXXXX).
- **Validation Rules:** FIN must be unique nationwide. Cell phone number verified via SMS OTP.
- **Priority:** Critical



FR-FARM-002: Geolocation Farm Polygon Capture (EUDR Ready)

- **Purpose:** Capture spatial polygon coordinates for every coffee farm plot to satisfy EUDR mandates.
- **Actors:** Field Inspector, Coop Agent.
- **Inputs:** GPS Spatial Vertices (Minimum 6 coordinate pairs for plots > 4,000 sqm; single point radius for plots < 4,000 sqm), Altitude, Primary Coffee Varieties, Estimated Tree Count.
- **Outputs:** Valid GeoJSON Polygon object with calculated land area (hectares) and spatial topology validation.
- **Business Rules:** Plots overlapping designated national forest reserves or deforested zones (post-December 31, 2020 baseline) are automatically flagged as non-EU export compliant.
- **Priority:** Critical

=====

MODULE 03: CHERRY COLLECTION & PRIMARY PROCESSING (FR-COLL)

=====

FR-COLL-001: Digital Weigh-Scale Intake & Receipt Issue

- **Purpose:** Record raw red cherry deliveries at primary collection centers directly from Bluetooth scales.
- **Actors:** Collection Agent, Farmer.
- **Inputs:** Farmer FIN, Cherry Type (Red Cherry / Floaters), Gross Weight (kg), Tare Weight (kg), Moisture Content (%), Visual Quality Grade.
- **Outputs:** Digital Collection Transaction, Physical Thermal Receipt, Automated SMS Notification to Farmer.
- **Business Rules:** Net Weight = Gross Weight - Tare Weight. System must reject moisture content readings > 65% for red cherries.
- **Validation Rules:** Bluetooth scale serial number must match registered center device to prevent manual entry tampering.
- **Priority:** Critical

FR-COLL-002: Wet Mill Batch Aggregation

- **Purpose:** Aggregate daily cherry collections into processing vats/lots.
- **Actors:** Cooperative Manager.
- **Inputs:** Collection Transaction IDs, Processing Method (Washed / Natural / Anaerobic), Fermentation Tank ID, Water Source ID.
- **Outputs:** Wet Batch Identification Number (WBIN - e.g., [WB-2026-SID-0042](#)).
- **Priority:** High

=====

MODULE 04: DRY MILLING & WAREHOUSE INVENTORY (FR-MILL)

=====

FR-MILL-001: Parchment Intake & Hulling Conversion

- **Purpose:** Track parchment coffee arrivals at dry mills and record green coffee hulling conversion ratios.
- **Actors:** Dry Mill Operator.

- **Inputs:** Incoming e-Waybill ID, Parchment Weight (kg), Moisture Content (%), Post-Hulling Green Coffee Weight (kg), Husk Weight (kg).
- **Outputs:** Green Coffee Lot ID, Conversion Efficiency Index (Target: 80-82% parchment-to-green yield).
- **Business Rules:** System generates an alert if weight loss exceeds 22% during hulling operations.
- **Priority:** Critical

FR-MILL-002: Export Bagging & Serialization

- **Purpose:** Track green coffee packed into standard 60kg export jute bags with grain-pro liners.
- **Actors:** Warehouse Supervisor.
- **Inputs:** Green Coffee Lot ID, Total Net Weight, Bag Count, Warehouse Zone/Aisle/Racking Slot.
- **Outputs:** Unique Serialized Barcode Tag applied to every individual 60kg bag.
- **Priority:** High

=====

MODULE 05: QUALITY INSPECTION & Q-GRADING (FR-QUAL)

=====

FR-QUAL-001: Anonymized Sample Blind Intake

- **Purpose:** Enforce double-blind quality grading to prevent corruption or bias.
- **Actors:** ECTA Quality Control Receptionist.
- **Inputs:** Incoming Lot GIN, Sample Weight (3 kg).
- **Outputs:** Blind Sample Code (BSC - Random 8-character alphanumeric string, e.g., X7K-9021).
- **Business Rules:** Unlinking between Lot GIN and BSC can only be performed by system workflow engine *after* Q-Graders submit final scores.
- **Priority:** Critical

FR-QUAL-002: Physical & Sensory Cupping Evaluation

- **Purpose:** Record physical defects and SCA sensory cupping parameters.
- **Actors:** Certified Q-Grader.
- **Inputs:** BSC, Primary Defects Count, Secondary Defects Count, Screen Size Distribution, Moisture Content %, Fragrance/Aroma (0-10), Flavor (0-10), Aftertaste (0-10), Acidity (0-10), Body (0-10), Balance (0-10), Uniformity (0-10), Clean Cup (0-10), Sweetness (0-10), Overall Score (0-10).
- **Outputs:** Cryptographically Signed ECTA Quality Grade Certificate (Grades 1-9 or Undergrade).

+-----+		
ECTA Q-GRADING SCORE MATRIX		
+-----+		
Total Score / Class	Assigned Grade	Value Chain Routing
+-----+		
85.00 - 100.00	Grade 1 (Specialty)	Premium Specialty Export
80.00 - 84.75	Grade 2 (Specialty)	Standard Specialty Export

70.00 - 79.75	Grade 3 (Commercial Commercial Export	
60.00 - 69.75	Grade 4 (Commercial Commercial Export	
50.00 - 59.75	Grade 5 (Commercial Commercial Export / Local Auction	
< 50.00 / High Def	Undergrade (UG) Local Domestic Market Only	
+-----+-----+-----+		

- **Priority:** Critical

=====

MODULE 06: TRACEABILITY & DYNAMIC QR ENGINE (FR-TRACE)

=====

FR-TRACE-001: Directed Acyclic Graph (DAG) Traceability Engine

- **Purpose:** Construct an unalterable lineage graph linking exported coffee bags back to farm origins.
- **Actors:** System Traceability Engine.
- **Inputs:** Processing Lot Split/Merge events, Bag Serialization Numbers, Cooperative Intakes.
- **Outputs:** Lineage Tree Payload (JSON-LD format) mapping % composition from each smallholder plot.
- **Priority:** Critical

FR-TRACE-002: Cryptographic QR Code Generation

- **Purpose:** Generate high-density 2D QR codes for bags, pallets, and export containers.
- **Actors:** System QR Generator.
- **Inputs:** GIN, Grade Certificate ID, EUDR Geolocation Hash, Exporter License Number.
- **Outputs:** Dynamic QR Code Image (PNG/SVG) containing a cryptographically signed HMAC-SHA256 URL
([https://trace.ecta.gov.et/v1/verify?id=\]\(https://trace.ecta.gov.et/v1/verify?id=\)...](https://trace.ecta.gov.et/v1/verify?id=](https://trace.ecta.gov.et/v1/verify?id=)...)).
- **Priority:** Critical

=====

MODULE 07: DIGITAL WAYBILL & GPS LOGISTICS (FR-LOG)

=====

FR-LOG-001: Electronic Waybill (e-Waybill) Issuance

- **Purpose:** Authorize legal transit of coffee across regional and national road networks.
- **Actors:** ECTA Regional Transport Officer.
- **Inputs:** Lot GIN, Bag Count, Weight, Dispatch Origin, Destination Warehouse ID, Truck Plate Number, Trailer Plate Number, Driver ID, Mobile Phone Number.
- **Outputs:** Digital Waybill Document (PDF + Active QR) with unique ID ([EWB-2026-XXXXXX](#)).
- **Business Rules:** Trucks cannot receive an e-Waybill if total gross weight exceeds Ethiopian Transport Authority axle load limits.
- **Priority:** Critical

FR-LOG-002: Geofenced GPS Transit Telematics

- **Purpose:** Track real-time movement of coffee trucks and detect unauthorized stops or route deviations.
- **Actors:** GPS Telematics Engine, Logistics Monitor.
- **Inputs:** Vehicle GPS coordinate stream (10-second intervals), Geofenced Corridor Boundaries.
- **Outputs:** Real-time route map display, Automated Tampering / Geofence Breach Alerts.
- **Business Rules:** Stopping outside designated police/inspection checkpoints for > 30 minutes triggers an immediate illegal offloading investigation alert.
- **Priority:** High

=====

MODULE 08: EXPORT LICENSING & FOREX TRACKING (FR-EXP)

=====

FR-EXP-001: Export Contract Registration & Minimum Price Check

- **Purpose:** Register international sales contracts and prevent under-invoicing.
- **Actors:** Exporter, ECTA Export Officer.
- **Inputs:** Buyer Name, Foreign Country, Coffee Variety, Grade, Volume (Metric Tons), Contract Price per lb (USD), Incoterms (FOB/CIF), Shipment Window.
- **Outputs:** Approved Export Contract Number (**EC-2026-XXXX**).
- **Business Rules:** Contract Price must be greater than or equal to ECTA's daily established Minimum Export Price for the specific coffee grade.
- **Priority:** Critical

FR-EXP-002: NBE Forex & Bank LC Validation Integration

- **Purpose:** Verify Foreign Exchange allocation and Commercial Bank Financial Guarantees.
- **Actors:** Bank Officer, NBE Automated Service.
- **Inputs:** Export Contract Number, Commercial Bank LC Reference Number, Swift Copy details, Escrow Account ID.
- **Outputs:** Automated Forex Verification Seal (**NBE-FX-APPROVED**).
- **Priority:** Critical

=====

MODULE 09: CUSTOMS CLEARANCE & PORT SHIPMENT (FR-CUST)

=====

FR-CUST-001: Container RFID / High-Security Seal Logging

- **Purpose:** Bind stuffed export containers to verified quality lots and digital permits.
- **Actors:** ECTA Port Inspector, Customs Officer.
- **Inputs:** Shipping Container Number, RFID Electronic Seal ID, Lot GINs, Export Permit Number.
- **Outputs:** Sealed Container Master Record.
- **Priority:** High

FR-CUST-002: Customs Exit Clearance & Port Intake

- **Purpose:** Execute final exit release at border posts (e.g., Galafi / Djibouti).
- **Actors:** Customs Officer.
- **Inputs:** Scanned e-Waybill QR Code, RFID Seal Scan.
- **Outputs:** Exit Clearance Approval, Automatic NBE Export Execution Notification.
- **Priority:** Critical

=====

MODULE 10: OFFLINE SYNCHRONIZATION ENGINE (FR-SYNC)

=====

FR-SYNC-001: Bidirectional Delta Synchronization

- **Purpose:** Synchronize offline mobile transactions with central cloud servers when connectivity becomes available.
- **Actors:** Android Offline Client, Cloud Sync Gateway.
- **Inputs:** Local SQLite Transaction Log queue, Client Vector Clock.
- **Outputs:** Sync Status Report, Server Ack Hashes, Conflict Resolutions.
- **Business Rules:** First-Writer-Wins policy for static data; Transaction-Log-Append for collection weights (no updates, append-only).
- **Priority:** Critical

4. External Interface Requirements

4.1 User Interfaces

- **Web Desktop Application:** Built with Angular 17 / Tailwind CSS. Optimized for 1024x768 resolution minimum, supporting dual-monitor setups for administrative officers. Must support native Amharic font rendering (Abyssinica SIL).
- **Mobile Field Application:** Built with Native Android (Kotlin). Optimized for rugged handheld tablets (7-inch and 10-inch screens) with minimum 1280x720 display resolution, readable under direct Sunlight (800+ nits display target).

4.2 Hardware Interfaces

- **Bluetooth Digital Weighing Scales:** Support RS-232 over Bluetooth SPP / BLE protocol for real-time auto-reading of weight values without human manual input.
- **Handheld 2D QR Barcode Scanners:** USB/Bluetooth HID profile support for Honeywell and Zebra industrial scanners.
- **Mobile Thermal Receipt Printers:** ESC/POS protocol support over Bluetooth BLE for field receipt issuing.

4.3 Software Interfaces

- **Database Interface:** PostgreSQL 16 connection via TLS 1.3 encrypted connection pools (PgBouncer).
- **Mapping Service Interface:** Mapbox Vector Tiles / OpenStreetMap hosted locally on EAI infrastructure for spatial rendering without internet dependency.

4.4 Communications Interfaces

- **HTTPS/REST API:** TLS 1.3 mandatory for all network traffic over JSON payloads.

- **SMS Gateway Interface:** SMPP v3.4 protocol connection to Ethio Telecom Short Message Service Center (SMSC).
- **USSD Gateway:** SIGTRAN / M3UA protocol integration for high-speed USSD session processing.

5. Non-Functional Requirements

NON-FUNCTIONAL REQUIREMENTS	
Metric / Category	Explicit System Target
Response Time	< 300 ms for 95% of API requests under normal load.
Peak Throughput	5,000 Concurrent Transactions / Second at harvest peak.
System Availability	99.95% Annual Uptime (Max 4.38 hours downtime/year).
Offline Retention	14 Days of operational transaction queuing on tablets.
Sync Execution Speed	< 15 seconds to sync 500 queued offline transactions.
RPO / RTO	Recovery Point Obj: 0 sec (sync) / Recovery Time Obj: 15m

5.1 Performance Requirements

- **NFR-PERF-01:** Web dashboard widgets must render fully within **1.5 seconds** on a standard 10 Mbps broadband line.
- **NFR-PERF-02:** Database query response times for complex traceability trees spanning 5,000 nodes must execute in **under 500 ms**.

5.2 Availability & Reliability

- **NFR-Avail-01:** Active-Active Multi-Zone Deployment across primary (Addis Ababa) and secondary (Bishoftu) cloud data centers.
- **NFR-RELI-01:** Mean Time Between Failures (MTBF) must exceed **2,160 hours**. Mean Time To Repair (MTTR) must be **less than 15 minutes**.

5.3 Usability & Accessibility

- **NFR-USE-01:** Mobile field interfaces must utilize high-contrast icons and visual indicators allowing semi-literate operators to complete intake workflows with **under 2 hours of formal training**.
- **NFR-USE-02:** Full localization support for English, Amharic, Afaan Oromo, and Tigrinya toggleable with one tap.

5.4 Maintainability & Portability

- **NFR-MAINT-01:** 100% modular microservices architecture utilizing Docker containers orchestrated via Kubernetes (K8s).
- **NFR-PORT-01:** Mobile applications must run on any standard Android device running Android 10.0 or higher.

6. Security Architecture & Requirements

SECURITY ARCHITECTURE	
Layer	Security Implementation
Transit Encryption	TLS 1.3 with AES-256-GCM cipher suites.
Storage Encryption	AES-256-XTS for databases and mobile SQLite DBs.
Identity Protocol	OAuth 2.0 with OpenID Connect (OIDC).
Access Model	Role-Based & Attribute-Based Access Control.
Integrity Protection	SHA-256 HMAC cryptographic transaction signatures

6.1 Authentication & Authorization

- **SEC-01:** Implement Role-Based Access Control (RBAC) combined with Attribute-Based Access Control (ABAC) to enforce regional data partitioning (e.g., Sidama region officers cannot modify Oromia region records).
- **SEC-02:** JWT Tokens must be signed using RS256 (RSA Signature with SHA-256) private keys held securely within a Hardware Security Module (HSM).

6.2 Encryption Standards

- **SEC-03:** All sensitive PII data (farmer phone numbers, bank details, National IDs) stored in the database must be encrypted at column level using AES-256 cipher keys.
- **SEC-04:** Mobile SQLite databases on Android tablets must be encrypted using SQLCipher with a key derived from device-specific hardware keystores (Android Keystore System).

6.3 Audit Trail & Immutable Logging

- **SEC-05:** Every CREATE, READ, UPDATE, DELETE (CRUD) action on commercial export records must be recorded in an append-only Audit Table storing: Timestamp (UTC), User ID, IP Address, Device Fingerprint, Old Value, and New Value.

7. Database Requirements & Data Architecture

7.1 Entity Relationship Diagram Description

The CEVCMS database architecture is modeled in PostgreSQL 16 with PostGIS spatial extensions. The core data models and relationships are defined below:

tbl_farmers	tbl_farms	tbl_collections
PK farmer_id	PK farm_id	PK collection_id
fin_code	FK farmer_id	FK farm_id
full_name	polygon_geom	weight_kg
phone_no	area_hectares	moisture_pct

			N
			V
+-----+	+-----+	+-----+	
tbl_e_waybills	tbl_quality_certs	tbl_batches	
+-----+	+-----+	+-----+	
PK waybill_id	PK cert_id	PK batch_id	
FK batch_id	<-----1 FK batch_id	<-----1 FK collection_id	
truck_plate	cup_score	wbin_code	
driver_id	assigned_grade	process_type	
+-----+	+-----+	+-----+	

7.2 Complete Data Schema Definitions

Table: `tbl_farmers`

SQL

```
CREATE TABLE tbl_farmers (
  farmer_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  fin_code VARCHAR(20) UNIQUE NOT NULL,
  full_name VARCHAR(150) NOT NULL,
  national_id VARCHAR(50) UNIQUE NOT NULL,
  gender CHAR(1) CHECK (gender IN ('M', 'F')),
  primary_coop_id UUID NOT NULL REFERENCES tbl_cooperatives(coop_id),
  phone_number VARCHAR(15) NOT NULL,
  payout_account_no VARCHAR(34),
  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_farmer_fin ON tbl_farmers(fin_code);
```

Table: `tbl_farms` (PostGIS Spatial Enabled)

SQL

```
CREATE TABLE tbl_farms (
  farm_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  farmer_id UUID NOT NULL REFERENCES tbl_farmers(farmer_id),
  farm_polygon GEOMETRY(Polygon, 4326) NOT NULL,
  altitude_meters NUMERIC(6,2),
  area_hectares NUMERIC(8,4) NOT NULL,
  eudr_deforestation_risk_flag BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_farms_spatial ON tbl_farms USING GIST(farm_polygon);
```

Table: `tbl_collections`

SQL

```
CREATE TABLE tbl_collections (
  collection_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  receipt_no VARCHAR(30) UNIQUE NOT NULL,
```

```

farm_id UUID NOT NULL REFERENCES tbl_farms(farm_id),
agent_id UUID NOT NULL REFERENCES tbl_users(user_id),
gross_weight_kg NUMERIC(8,2) NOT NULL,
tare_weight_kg NUMERIC(6,2) NOT NULL,
net_weight_kg NUMERIC(8,2) GENERATED ALWAYS AS (gross_weight_kg -
tare_weight_kg) STORED,
moisture_percentage NUMERIC(4,2) NOT NULL,
cherry_type VARCHAR(20) CHECK (cherry_type IN ('RED_CHERRY', 'FLOATERS')),
payout_amount_etb NUMERIC(10,2) NOT NULL,
offline_created_at TIMESTAMP WITH TIME ZONE NOT NULL,
synced_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

8. API Specifications & Webhooks

CEVCMS exposes RESTful APIs adhering to OpenAPI 3.0 standards. All API request and response bodies use standard UTF-8 encoded JSON format.

8.1 API Endpoints Specification

1. POST [/api/v1/collections/intake](#)

- **Description:** Submit a new cherry collection transaction from field tablets.
- **Headers:** [Authorization: Bearer <JWT>](#), [Content-Type: application/json](#)
- **Request Payload:**

JSON

```

{
  "offline_transaction_id": "8f3c1a2b-4e5d-6c7b-8a9b-0c1d2e3f4a5b",
  "farmer_fin": "ETH-FAR-SID-883921",
  "farm_id": "c1a2b3c4-d5e6-7f8a-9b0c-1d2e3f4a5b6c",
  "scale_device_mac": "00:11:22:33:FF:EE",
  "gross_weight": 142.50,
  "tare_weight": 2.50,
  "moisture_pct": 55.40,
  "cherry_type": "RED_CHERRY",
  "timestamp_utc": "2026-08-07T13:45:00Z"
}

```

- **Response Payload (201 Created):**

JSON

```

{
  "status": "SUCCESS",
  "server_receipt_no": "REC-2026-SID-094821",
  "collection_id": "d9e8f7c6-b5a4-3f2e-1d0c-9b8a7f6e5d4c",
  "payout_status": "QUEUED_TELEBIRR",
  "signature_hash": "a5d8f9e...3b1c"
}

```

2. GET /api/v1/traceability/verify/{gin}

- **Description:** Public endpoint to retrieve cryptographic origin and EUDR compliance data for a specific lot.
- **Response Payload (200 OK):**

JSON

```
{
  "gin": "ETH-LOT-2026-G1-00392",
  "quality_grade": "Grade 1 - Specialty",
  "cupping_score": 87.50,
  "origin_region": "Yirgacheffe, Sidama",
  "cooperative": "Yirgacheffe Union Mill #2",
  "farm_count": 42,
  "eudr_compliant": true,
  "deforestation_free_hash":
    "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855",
  "processing_type": "Washed",
  "export_contract_id": "EC-2026-8812"
}
```

9. Business Rules Engine

BUSINESS RULES LISTING	
Rule ID	Rule Description & Technical Logic Enforcement
BR-001	Minimum Contract Price Validation: Contract Price >= ECTA Daily Floor Price.
BR-002	Dynamic Weight Loss Threshold: Hulling Parchment Weight Loss capped at 22%.
BR-003	Waybill Transit Expiry: e-Waybills automatically expire after 72 hours.
BR-004	Double-Cupping Mandate: Samples with score >= 88.00 require second master cup
BR-005	Forex Lock Rule: Export License cannot be issued without valid NBE Forex Ref.
BR-006	EUDR Strict Lock: Lots containing unmapped farms CANNOT be added to EU containers.
BR-007	Moisture Limit Rule: Green coffee moisture MUST be between 9.5% and 11.5%.

10. Comprehensive Use Cases

This section documents **40 major use cases** covering the entire coffee export value chain.

USE CASE DIRECTORY (40 CORE CASES)	
UC-ID	Title

+-----+-----+-----+-----+-----+	
UC-01 Register Smallholder Farmer	UC-21 Issue Quality Grade Certificate
UC-02 Capture GPS Farm Polygon	UC-22 Split Coffee Processing Lot
UC-03 Log Red Cherry Collection	UC-23 Merge Green Coffee Lots
UC-04 Aggregate Wet Mill Batch	UC-24 Apply Serialized Bag Barcodes
UC-05 Print Thermal Field Receipt	UC-25 Export Contract Application
UC-06 Payout via Mobile Money	UC-26 NBE Minimum Price Check
UC-07 Register Parchment Transit	UC-27 Commercial Bank LC Verification
UC-08 Issue Parchment e-Waybill	UC-28 Generate ECTA Export Permit
UC-09 Inspect Parchment Checkpoint	UC-29 Assign Shipping Container Seals
UC-10 Intake Dry Mill Lot	UC-30 Issue Export e-Waybill
UC-11 Hull Parchment to Green	UC-31 Activate GPS Telematics Transit
UC-12 Record Hulling Loss Yield	UC-32 Detect Geofence Route Breach
UC-13 Store Warehouse Pallet	UC-33 Customs Border Exit Release
UC-14 Receive Grading Sample	UC-34 Confirm Foreign Buyer Receipt
UC-15 Assign Blind Sample Code	UC-35 Perform Delta Sync (Offline)
UC-16 Perform Screen Size Sorting	UC-36 Resolve Sync Conflict
UC-17 Perform Physical Defect Count	UC-37 Broadcast Daily Market Prices
UC-18 Conduct Blind Cupping Test	UC-38 Generate Executive Report
UC-19 Calculate Final Cup Score	UC-39 Audit System Action Logs
UC-20 Re-Cup Disputed Score	UC-40 Configure User RBAC Permissions
+-----+-----+-----+-----+-----+	

Detailed Specifications for Representative Use Cases

USE CASE UC-01: Register Smallholder Farmer

- **Primary Actor:** Cooperative Agent / Field Inspector.
- **Preconditions:** Agent is authenticated on mobile tablet; tablet has active GPS signal.
- **Main Success Scenario:**
 1. Agent enters farmer's Personal Details (Name, National ID, Phone, Gender).
 2. System validates uniqueness of National ID and Phone Number.
 3. Agent captures photo of National ID document.
 4. Agent selects primary cooperative and enters financial payout details.
 5. System generates FIN (**ETH-FAR-XXX-XXXXXX**), signs transaction cryptographically, and outputs digital registration confirmation.
- **Alternative Flow (Offline):** Transaction is queued in local SQLite database; FIN is assigned using offline local cryptographic hash block.
- **Postconditions:** Farmer profile is saved and ready for farm polygon binding.

USE CASE UC-02: Capture GPS Farm Polygon (EUDR)

- **Primary Actor:** Field Inspector.
- **Preconditions:** Farmer is registered in system; Inspector is physically standing on plot boundary.
- **Main Success Scenario:**
 1. Inspector initiates polygon capture on mobile tablet.
 2. Inspector walks perimeter of plot; GPS logs spatial vertices at every direction change (min 6 points).

3. Inspector completes perimeter walk; system closes polygon loop.
 4. System calculates total acreage and executes PostGIS spatial query against deforestation forest baseline map.
 5. System flags plot as **EUDR_COMPLIANT_PASS** and attaches GeoJSON polygon to Farmer Profile.
- **Exception Flow (Deforestation Detected):** System flags plot as **EUDR_RISK_FAIL**. Issue warning alert to inspector; block future lot exports from this plot to EU markets.

USE CASE UC-18: Conduct Blind Cupping Test

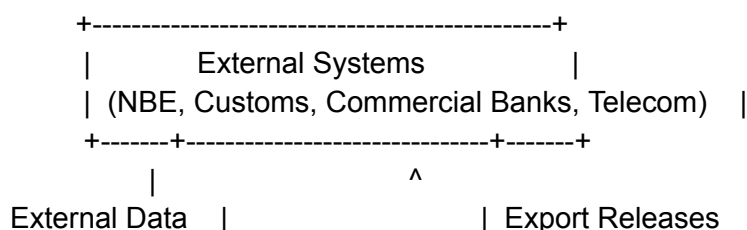
- **Primary Actor:** Certified Q-Grader.
- **Preconditions:** Blind Sample Code (BSC) has been prepared and assigned to cupping tray.
- **Main Success Scenario:**
 1. Q-Grader enters or scans BSC on touch terminal.
 2. Q-Grader enters individual sensory attributes (Fragrance, Flavor, Acidity, Body, Clean Cup, Uniformity, Sweetness).
 3. System automatically sums sensory scores and subtracts physical defect deductions.
 4. Q-Grader clicks "Submit Cupping Score".
 5. System verifies completeness and logs score with Q-Grader's digital cryptographic signature.
- **Postconditions:** Sample score is logged under BSC; system evaluates if second cup is required (Rule BR-004).

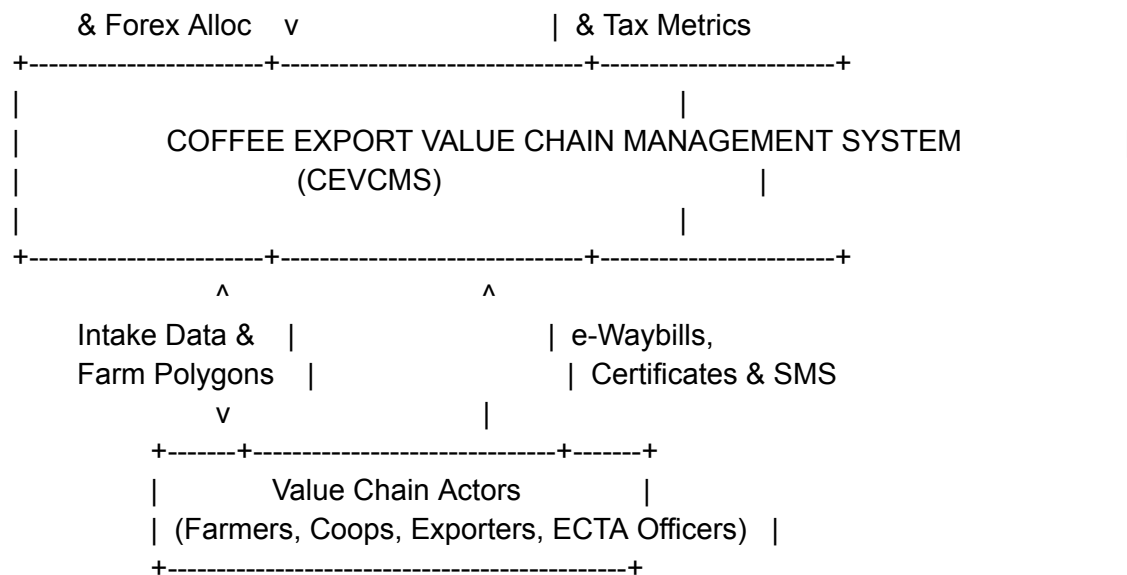
USE CASE UC-30: Issue Export e-Waybill

- **Primary Actor:** ECTA Transport Officer.
- **Preconditions:** Export License approved; Quality Grade Certificate issued; Truck Plate and Driver verified.
- **Main Success Scenario:**
 1. Officer inputs Export Contract ID and Lot GINs.
 2. System validates that container seals are intact and match warehouse dispatch logs.
 3. Officer inputs Truck Plate, Trailer Plate, Driver License ID, and Destination Port.
 4. System generates **EWB-2026-XXXXXX** e-Waybill document containing dynamic QR code and signs payload with ECTA private key.
 5. System transmits e-Waybill copy to Driver Mobile App and Customs Exit Gate Gateway.

11. Data Flow Diagrams (DFD)

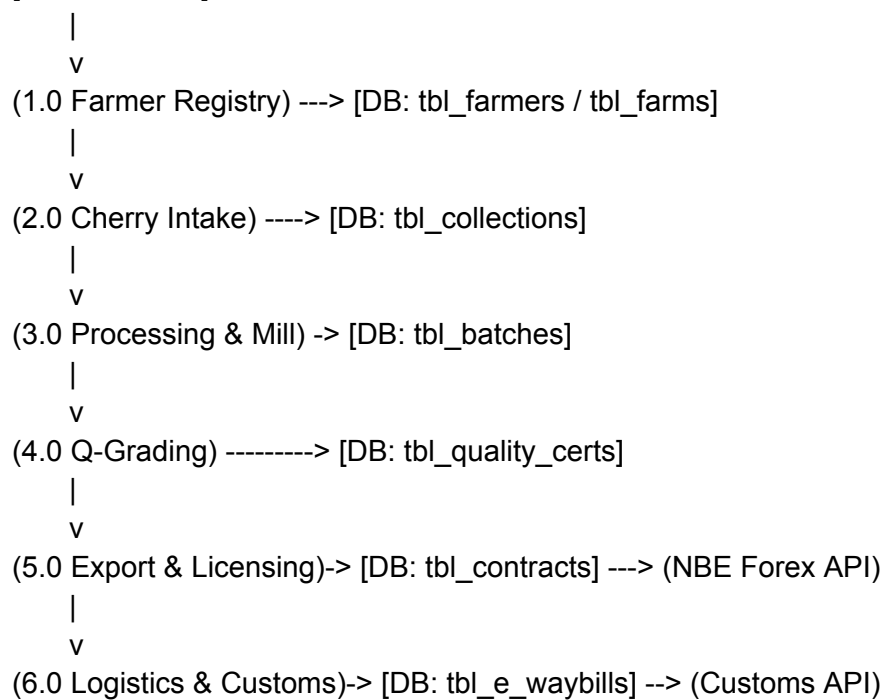
11.1 Level 0 Context Diagram





11.2 Level 1 Process Diagram

[Farmer & Plot]



12. Risk Analysis & Mitigation Strategies

Risk ID	Category	Risk Description	Severity	Impact	Mitigation Strategy

RSK-01	Infrastructure	Rural cellular network blackout lasting > 7 days during peak harvest season.	High	High	Implement offline CouchDB engine with SQLite local transaction queuing capable of 14 days storage.
RSK-02	Fraud	Collection agents entering fake cherry weights to siphon cooperative funds.	Critical	Critical	Direct Bluetooth integration with scales; enforce physical check weights and strict weight-loss ratio alerts.
RSK-03	Compliance	EUDR rejection of export containers due to inaccurate farm boundary mapping.	Critical	High	Mandatory PostGIS polygon topology validation; automated cross-check against 2020 Global Forest Canopy maps.
RSK-04	Security	Interception and falsification of digital e-Waybill QR codes by illegal smugglers.	High	Critical	Cryptographic HMAC-SHA256 signature embedded in dynamic QR code verified against ECTA cloud public key.

RSK-05	Adoption	Resistance from traditional dry mill operators and exporters to new software.	Medium	Medium	Executive directive mandating digital permits for customs clearance; intensive regional training programs.
--------	----------	---	--------	--------	--

13. Testing & Quality Assurance Strategy

TESTING PIPELINE MATRIX			
Test Level	Target Scope	Automation Tooling	
Unit Testing	Microservice business logic code	PyTest / JUnit 5 / Jest	
Integration Test	REST API contracts & DB queries	Postman / Testcontainers	
System E2E	End-to-end value chain workflows	Playwright / Cypress	
Mobile QA	Android offline sync & Bluetooth	Appium / Android Runner	
Performance Load	Peak harvest API load (5k TPS)	Apache JMeter / Locust	
Security Audit	OWASP Top 10 & Pen Testing	SonarQube / OWASP ZAP	

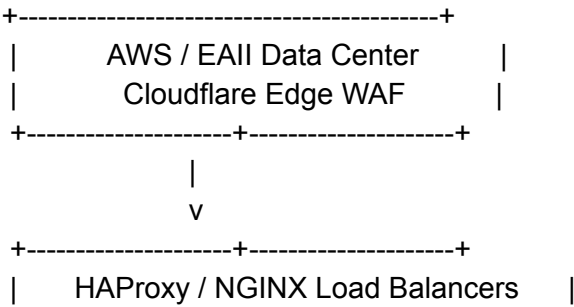
13.1 Offline Sync Validation Protocol

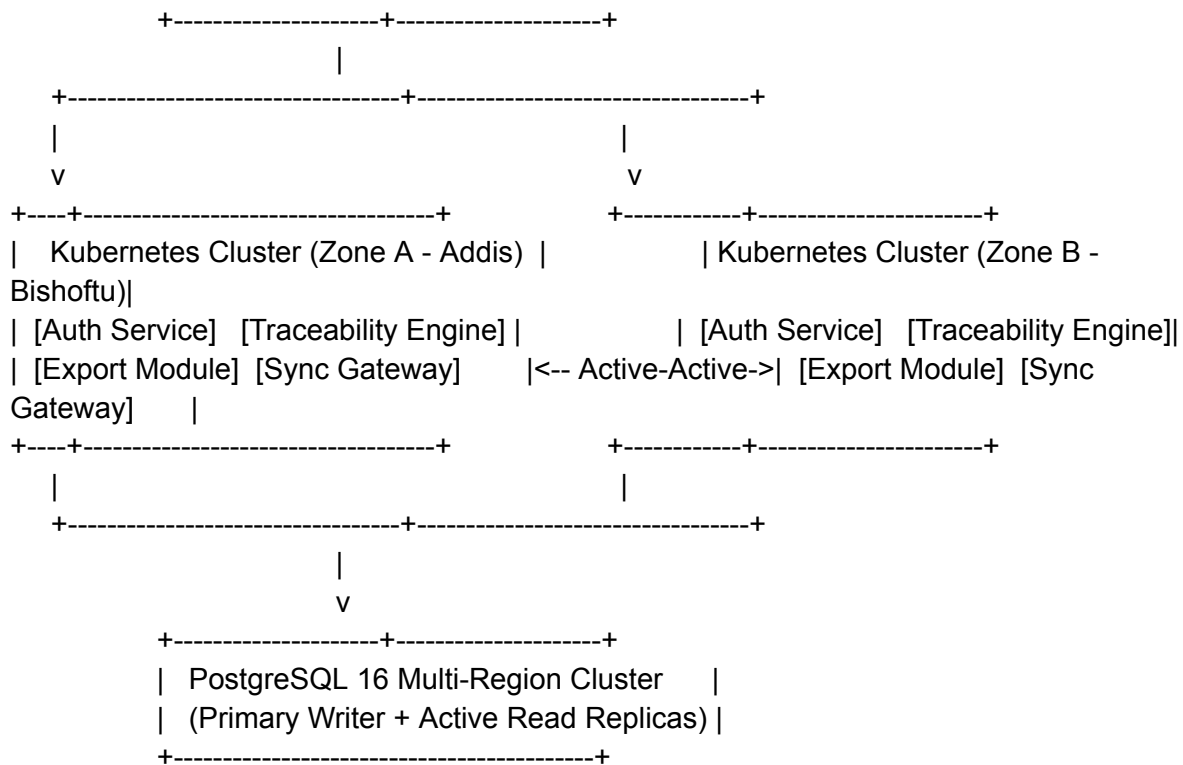
Specific offline testing requires placing test tablets into Faraday cages to simulate total network isolation:

- 1. Execute 2,000 collection transactions on isolated device.
- 2. Simulate power loss and hard restart device.
- 3. Restore device network connection; verify zero data loss and sync completion within 30 seconds.

14. Deployment Architecture & Infrastructure

14.1 Production Cloud Infrastructure Layout





14.2 Hardware Specifications

Primary Application Servers (Kubernetes Nodes)

- **Quantity:** 6 Nodes (3 per Availability Zone).
- **Specification:** 32 vCPU, 128 GB RAM, 1 TB NVMe SSD, Dual 10 Gbps Network Interface Cards.

Database Server Nodes (Bare Metal)

- **Quantity:** 3 Nodes (1 Primary, 2 Synchronous Standbys).
- **Specification:** 64 Core CPU, 512 GB RAM, 4 TB NVMe RAID-10 Storage array.

15. Future Enhancements & EUDR Roadmap

FUTURE ENHANCEMENT ROADMAP	
Phase	Initiative Description
Phase 3 (2027)	AI Computer Vision Green Coffee Defect Analysis.
Phase 3 (2027)	Satellite Remote Sensing for Automatic Yield Estimation.
Phase 4 (2028)	Distributed Ledger / Blockchain Origin Smart Contracts.
Phase 4 (2028)	Farmer Digital Credit Scoring based on historical yield.
Phase 5 (2028+)	IoT Automated Moisture & Temperature Warehouse Sensors.

16. Appendices

Appendix A: System Acronyms

- **API:** Application Programming Interface
- **EWB:** Electronic Waybill
- **FIN:** Farmer Identification Number
- **GIN:** Global Identification Number
- **NBE:** National Bank of Ethiopia
- **Q-Grading:** Quality Grading
- **WBIN:** Wet Batch Identification Number

Appendix B: Requirement Traceability Matrix (RTM)

Requirement ID	Functional Module	High-Level Business Goal	Test Case ID
FR-FARM-001	Farmer Registry	Smallholder Identification	TC-FARM-01
FR-FARM-002	Polygon Capture	EUDR Deforestation Compliance	TC-FARM-02
FR-COLL-001	Cherry Intake	Digital Weight Verification	TC-COLL-01
FR-QUAL-002	Q-Grading	Quality Certification	TC-QUAL-01
FR-EXP-001	Export Licensing	Foreign Exchange Protection	TC-EXP-01
FR-SYNC-001	Offline Sync Engine	Rural Operational Continuity	TC-SYNC-01

Appendix C: Cryptographic e-Waybill Verification Layout

+-----+		
	ETHIOPIAN COFFEE AND TEA AUTHORITY (ECTA)	
	OFFICIAL DIGITAL E-WAYBILL	
+-----+		

WAYBILL NO	: EWB-2026-SID-009412	DISPATCH DATE	: 2026-08-07
BATCH GIN	: ETH-LOT-2026-G1-00392	BAG COUNT	: 320 Bags
WEIGHT	: 19,200.00 KG (Net)	GRADE	: Grade 1 Specialty
+-----+			
ORIGIN	: Yirgacheffe Union Mill #2, Sidama Zone		
DESTINATION	: ECTA Export Terminal #1, Akaki-Kality, Addis Ababa		
TRUCK NO	: ETH 3-A-89212	DRIVER LIC	: DL-ETH-901284
+-----+			
	[DYNAMIC Cryptographic QR Code]		
	HMAC-SHA256 Signed by ECTA HSM		
+-----+			
VERIFICATION URL	: https://trace.ecta.gov.et/v1/verify?id=EWB-009412		
+-----+			

[END OF SOFTWARE REQUIREMENTS SPECIFICATION — ECTA CEVCMS V2.1]